

The Dog That Didn't Bark: Catching What Stops Arriving, While the Stream Runs

Technical Report

This report carries the statements of Secs. 5.2–5.4 of the paper with their proofs in full, the paper's Prop. 5.1 in the form the proofs use, its invariant (I2); the LETHE-B results of Sec. 5.5 are proved in the paper itself. Equations numbered (R1) onward are this report's; un-prefixed equation numbers are the paper's. The report's Lemma A.3 has no counterpart in the paper, so the paper's Thm. 5.3, Thm. 5.4 and Cor. 5.5 are A.4, A.5 and A.6 here.

A Notation, Statements and Proofs

This appendix gives the inventory of writes the proofs rely on, the statements of Sec. 5 of the paper restated for reference, and the proofs, which use (3), (4) and Alg. 1 of the paper.

Writes and invariants. Every write to a bucket comes from one of

$$\mathcal{P} = \{\text{INSTALL, HIT, DRAIN, SCAN}\},$$

where HIT covers Lines 5–7 and DRAIN is the on-miss decrement of Lines 16–17. INSTALL sets $\hat{P}_b \leftarrow 1$ and writes only inside the installing item's own probe set,

$$\text{INSTALL}(b, f) \text{ occurs} \implies b \in B(f); \quad (\text{R1})$$

HIT raises the count at Line 7, at most once per window, and DRAIN and SCAN lower it at Lines 17 and 23. Two writers leave τ_b alone, DRAIN and the release branch of SCAN at Line 21. Neither can change id_b , and while $\hat{P}_b > 0$ the last arrival of id_b found b at Lines 1–2 and wrote its own window, so, writing $\pi_c(f) = \max(\{0\} \cup \{t < c : a_f(t) = 1\})$ for the last completed window in which f arrived,

$$\tau_b \geq \pi_c(\text{id}_b). \quad (\text{R2})$$

Three assumptions are carried throughout, as in Sec. 5.1 of the paper: (A0) is Alg. 1 with the scan complete at the close of each window; (A1) is that each h_i is a uniformly random function into $[M/d]$, the d of them independent of one another, of the stream and of the coins; (A2) is $L \geq 2$ and $1 \leq g < W$. Of the paper's invariants the proofs below use (I1), that at most one bucket carries f with a positive count, and (I5), that under (A0) an unreported bucket with a positive count has $s_c(b) \leq g-1$ once the scan of c has run, so that every report comes from Line 21. (I5) is the scan itself: a bucket with $\hat{P}_b > 0$, $r_b = 0$ and $s_c(b) \geq g$ is visited, and Line 21 leaves it reported, or Line 24 empties it or sets $\tau_b \leftarrow c$, so $s_c(b) = 0 \leq g-1$ by (A2); a bucket the scan passes over has $s_c(b) < g$. Also Prop. A.1 is (I2).

Comparing Lines 3, 12 and 19–20 with (3) collapses the three emission guards into one:

$$\text{a declaration leaves } b \text{ at } c \implies \text{due}_c(b). \quad (\text{R3})$$

Statements. The statements of Sec. 5 of the paper, restated for reference; Table 1 collects the notation.

Table 1: Notation used throughout.

<i>The stream and the event</i>	
T, W, N	arrivals; windows; arrivals per window
f, c	an item; the current window
$a_f(t)$	1 if f arrives in window t
$P_f, \hat{P}$	persistence; the count a bucket holds
t_{last}	the most recent window with $a_f = 1$
P_{eff}, γ	length and largest gap of the run before a vanish
L, α	establishment threshold $L = \alpha W$
g	silent windows confirming a vanish
t_d	declaration window
<i>The sketch</i>	
M, d	buckets; rows
$h_1, \dots, h_d$	the per-row hashes
τ, r	a bucket's last-seen field; reported flag
κ	decay constant of the scan
N_c, λ, q	distinct items; dN_c/M ; chance a bucket is shared
$s_c(b)$	silence of b at window c
A, A_i	a competitor probes every bucket of f ; the one in row i
n, z	drains from t_1 that leave the count positive; that reach zero
V	the event $n+z > P_{\text{eff}} - L$, on A^c

PROPOSITION A.1 (THE COUNT NEVER OVERESTIMATES; (I2) OF THE PAPER'S PROP. 5.1). *If $\text{id}_b = f$ and $\hat{P}_b > 0$ then $\hat{P}_b \leq P_f(c)$ at every window c , and $\hat{P}_b \leq P_f(c-1)$ whenever $\tau_b \neq c$.*

PROPOSITION A.2 (NO FALSE POSITIVES). *Let LETHE declare (f, c) at a window $c \leq W-1$, and put $t^* = \pi_c(f)$ as in (R2), the last completed window in which f arrived, a window that exists because the declaration needs $\hat{P}_b \geq L \geq 2$. Then*

$$P_f(t^*) \geq L, \quad a_f(t) = 0 \text{ for } t^* < t \leq c, \\ c - t^* = g,$$

so (f, t^*) is a vanish event, the declaration matches it, and no silence run yields two declarations.

LEMMA A.3 (PROTECTION). *$\text{est}(b) \wedge \neg \text{rep}(b) \wedge \neg \text{due}_c(b)$ implies that no member of $\mathcal{P}$ writes b at window c except HIT on id_b .*

THEOREM A.4 (EXACT DELAY). *Let (f, t_{last}) be a vanish event, and let b satisfy $\text{id}_b = f$, $\text{est}(b)$ and $\neg \text{rep}(b)$ at the end of window t_{last} . Then LETHE declares (f, t_d) with $t_d = t_{\text{last}} + g$.*

Setup for the miss bound. Fix a vanish event (f, t_{last}) with $t_{\text{last}} \leq W-1-g$, so that the declaration of Thm. A.4 falls inside the stream. Let

$$I = \{t_1 < \dots < t_m\}, \quad t_m = t_{\text{last}}, \quad \gamma = \max_{i < m} (t_{i+1} - t_i) \leq g$$

be the longest terminal run of arrival windows of f in which no gap exceeds g , with $\gamma = 1$ when $m = 1$, and put $P_{\text{eff}} = m$. A larger gap cuts the run, because a silence of g windows leaves the bucket

exposed: an established bucket is reported, and a climbing one meets the decay of Line 23 of Alg. 1.

Let N_c be the number of distinct items that arrive in windows $[0, t_{\text{last}}+g]$; every item resident in the table at t_1 arrived earlier, so the stream alone fixes the competitor set. Let

$$A_i = \{\text{another item probes } f\text{'s bucket in row } i\},$$

$$A = \bigcap_{i=1}^d A_i,$$

and write

$$q = 1 - \left(1 - \frac{d}{M}\right)^{N_c-1}, \quad \lambda = \frac{dN_c}{M},$$

so that $\Pr[A_i] = q$; λ is the load factor, the average number of probes a bucket receives over that interval. Writing $\lambda' = \lambda(N_c-1)/N_c$, Bernoulli's inequality and $1-x \leq e^{-x} \leq 1-x+x^2/2$ give

$$\lambda' - \frac{1}{2}\lambda'^2 \leq q \leq \min(1, \lambda') \quad \text{and} \quad q^d = \lambda'^d(1+O(\lambda')),$$

hence $q < \min(1, \lambda)$, and at a fixed $N_c \geq 2$, as M grows, $q^d \asymp \lambda^d$ with $\partial \log q^d / \partial \log \lambda \rightarrow d$.

Event A collects the misses in which a competitor holds every bucket f could use. The route that remains leaves f in its bucket and takes its count instead. On A^c , and only there, f holds a bucket at every arrival window of I from t_1 on (step (i) of the proof below), and from that arrival its count is written only by HIT, DRAIN and the INSTALL that follows a drain to zero, with HIT writing at most once per window of I . Counting from t_1 , and on that event, let n be the drains that leave the count positive and z those that reach zero, and let

$$V = \{n+z > P_{\text{eff}} - L\}$$

be the event that the drains exhaust the run's margin over L .

THEOREM A.5 (MISS PROBABILITY). *In the setting above,*

$$\Pr[\text{LETHE misses}(f, t_{\text{last}})] \leq q^d + \Pr[V \cap A^c].$$

If moreover $d \leq 2$, $\gamma \leq 2$ and $P_{\text{eff}} \geq L+1$, then $V \cap A^c$ is empty and the bound is q^d .

COROLLARY A.6 (ERROR OF THE REPORTED PERSISTENCE). *In the setting of Thm. A.5, on $A^c \cap V^c$, the bucket b holding f at t_{last} satisfies*

$$0 \leq P_f(t_{\text{last}}) - \widehat{P}_b|_{t_{\text{last}}} \leq P_f(t_{\text{last}}) - P_{\text{eff}} + n + z,$$

which is the paper's form, since $P_{\text{eff}} = P_f(t_{\text{last}}) - P_f(t_1-1)$.

Proofs. Statements are those of Sec. 5 and line numbers those of Alg. 1.

Proof of Prop. A.1. The count a bucket carries never exceeds its item's persistence. By Lines 1–2 an arrival of f tests $\text{id}_b = f$ over all of $B(f)$ before any placement, so with (R1) at most one bucket carries f with a positive count. Let c_0 be the window of the INSTALL that wrote $\text{id}_b = f$; that install runs on an arrival of f , so $a_f(c_0) = 1$. Of the writes to $\widehat{P}_b$ only HIT raises it, and it fires at c only under $\text{id}_b = f \wedge a_f(c) = 1 \wedge \tau_b \neq c$, hence at most once per window and only in windows where f arrives. Therefore

$$\widehat{P}_b \leq |\{t \in [c_0, c] : a_f(t) = 1\}| \leq P_f(c).$$

If $\tau_b \neq c$ then neither an INSTALL nor Line 7 ran on b in window c , both writing $\tau_b \leftarrow c$; every other write lowers the count, so $\widehat{P}_b$ is at most its value at the close of $c-1$, hence at most $P_f(c-1)$.

Proof of Prop. A.2. Every declaration names an established item and a genuine silence run, and arrives inside the delay window. By (R3) the declaration requires $\text{due}_c(b)$, hence $s_c(b) \geq g$ and so $\tau_b \neq c$, so b carries no HIT of window c ; Prop. A.1 then gives $P_f(c-1) \geq \widehat{P}_b \geq L$, and P_f does not move between t^* and $c-1$, so $P_f(t^*) \geq L$. The silence claim below c is the definition of $t^* = \pi_c(f)$. At c itself, (I5) makes Lines 4 and 13 unreachable under (A0), so the emission is Line 21's, at the close of c ; by (I1) b is f 's only bucket with a positive count, and an arrival of f in c would have left $\tau_b = c$ at Line 7, against $\tau_b \neq c$. Hence $a_f(c) = 0$.

For the lower end of the delay, (R2) gives $\tau_b \geq t^*$, so

$$s_c(b) = c - \tau_b \leq c - t^*,$$

and $s_c(b) \geq g$ forces $c - t^* \geq g$.

For the upper end, consider the bucket at the end of window t^* . It carries $\text{id} = f$, because a later install would reset $\widehat{P}_b$ to $1 < L$ by (A2); it is established, because $\widehat{P}_b$ only rises by Line 7 and the rise happens at arrivals of f , of which there are none after t^* ; and it is unreported, because r_b is set only at Line 21 and cleared only at Line 5 or by an install, which resets the count, and a set bit would block the declaration at c through (3). Thm. A.4 then gives $c - t^* = g$, the declaration it names being this one by the uniqueness below. So (f, t^*) is matched.

For uniqueness, the write that accompanies each emission falsifies a conjunct of (3) at that window: Line 4 is followed by $\tau_b \leftarrow c$ at Line 7, admissible because $s_c(b) \geq g$ forces $\tau_b \neq c$; Line 13 sets $\widehat{P}_b \leftarrow 1 < L$ by (A2); Line 21 sets $r_b \leftarrow 1$.

Proof of Lem. A.3. An established, unreported, not-yet-due bucket is written only by its own item. The three hypotheses put b in the second row of (4), so $\text{rank}_c(b) = W+1$, the maximum of the range, and a miss that selects b reaches Line 14 and skips at Line 15. Line 10 is excluded because b is neither empty nor reported, Line 12 because b is not due, and Lines 16–17 because $\text{est}(b)$ sends the test to Line 14 first. SCAN tests $\neg \text{rep}(b) \wedge s_c(b) \geq g$ at Line 19, which under $\text{est}(b)$ is $\text{due}_c(b)$, excluded. The only writer left is HIT, which runs on id_b .

Proof of Thm. A.4. The declaration lands exactly g windows after t_{last} .

Base. At the end of t_{last} we have $\tau_b = t_{\text{last}}$. By (I1) of Prop. 5.1 of the paper at most one bucket carries f , so f 's arrival in t_{last} took the Line 2 branch on b ; and b was not installed in t_{last} , since an install leaves $\widehat{P}_b = 1 < L$ by (A2) while $\text{est}(b)$ holds. So Line 7 ran and wrote $\tau_b \leftarrow t_{\text{last}}$, or found it there already.

Before the g th silent window closes. Suppose b is unwritten over $(t_{\text{last}}, c-1]$ for some $c < t_{\text{last}}+g$, so $\tau_b = t_{\text{last}}$ and $s_c(b) = c - t_{\text{last}} < g$, giving $\neg \text{due}_c(b)$. Def. 2.1 gives $a_f(c) = 0$, so Line 2 never matches b and HIT does not run, and r_b is unchanged because Line 19 also needs $s_c(b) \geq g$. Lem. A.3 leaves b unwritten at c , closing the induction.

At the close of $t_{\text{last}}+g$. Inside that window an arrival reads $s_c(b) = g$, which is not past g , so b is not due: a miss that selects it falls to Line 15 and skips, and HIT does not run since f does not arrive. When the window closes the same silence reads $s_c(b) \geq g$ and $\text{due}_c(b)$ holds: Line 19 admits b , Line 20 holds, so Line 21 emits

before it writes. The scan visits every bucket when the window closes and this is the first window at which b is due, so $t_d = t_{\text{last}} + g$.

Proof of Thm. A.5. $\Pr[\text{miss}] \leq q^d + \Pr[V \cap A^c]$. The rows are disjoint and (A1) makes the h_i uniform and independent, so

$$\Pr[A] = \prod_{i=1}^d \Pr[A_i] = q^d. \quad (\text{R4})$$

Assume A^c and fix $e \in B(f)$ with $e \notin B(f')$ for every other item f' that arrives in $[0, t_{\text{last}} + g]$, a set that contains every item resident at t_1 . By (R1) no INSTALL but f 's writes e , and neither HIT nor a miss reaches it, so

$$\text{emp}(e) \vee \text{id}_e = f \quad \text{at every window.} \quad (\text{R5})$$

(i) f holds a bucket from t_1 on. If f is not resident at $t_j \in I$ then $\text{emp}(e)$ by (R5), so $\min_{b \in B(f)} \text{rank}(b) = \text{rank}(e) = 0$ by (4). Every rank-0 bucket satisfies $\text{emp} \vee \text{rep} \vee \text{due}$, and Lines 10–13 install into all three, so f is installed at t_j whichever minimiser the tie-break returns.

(ii) Only DRAIN lowers the count. For b holding f from f 's arrival at t_1 on, $s_c(b) \leq \gamma \leq g$ at an arrival and $s_c(b) \leq \gamma - 1 < g$ at a close, so $\neg \text{due}_c(b)$; that excludes Line 12, while Line 10 is excluded by $\neg \text{emp}(b) \wedge \neg \text{rep}(b)$, the second conjunct because r is set only at Line 21, which needs $s_c(b) \geq g$, as does the decay of Lines 22–24, and Line 14 skips whenever $\text{est}(b)$. The one surviving path is Line 16, which requires $s_c(b) > \widehat{P}_b$ and hence $\widehat{P}_b < \gamma$.

(iii) At most one drain reaches zero, for $d \leq 2$. Let one do so in a window w ; Line 18 installs the arriving item at once, so $\text{rank}(y) = 1$. Before f 's next arrival y 's silence is at most $\gamma \leq g$ at an arrival and

$\gamma - 1$ at a close, so it cannot become due or reported, and cannot become empty, since Line 18 reinstalls on reaching zero and Line 24 needs $s \geq g$; so $\text{rank}(y) \geq 1 > 0$ while $\text{rank}(e) = 0$. At $d = 2$, $B(f) = \{e, y\}$, so f installs into e , which (R5) makes unreachable; at $d = 1$, $B(f) = \{e\}$ and no drain occurs. Hence $z \leq 1$ on A^c when $d \leq 2$; A is charged in full by (R4), so V is intersected with A^c .

(iv) The count reaches L . Both updates of (ii), $x \mapsto x+1$ and $x \mapsto x-1$, are non-decreasing and 1-Lipschitz, and a drain reaching zero is followed at f 's next arrival by an INSTALL writing $1 = 0+1$, the same map; so deleting a drain raises the final count by exactly one and never lowers it. Counting the writes of (ii) from f 's arrival at t_1 against the drain-free trajectory,

$$\widehat{P}_b|_{t_{\text{last}}} \geq P_{\text{eff}} - n - z. \quad (\text{R6})$$

On V^c the right-hand side is at least L , so the count reached L , and no path lowers it after that, Lines 14–15 skipping an established bucket and the decay of Lines 22–23 needing $\neg \text{est}$. A bucket at t_{last} therefore carries $\text{id} = f$, est and $\neg \text{rep}$, the last because SCAN never reached it, and Thm. A.4 declares (f, t_d) inside the matching window. Hence $\text{miss} \subseteq A \cup (V \cap A^c)$, which with (R4) is the bound.

For $d \leq 2$ and $\gamma \leq 2$ on A^c , a drain needs $\widehat{P}_b < s_c(b) \leq 2$, so $\widehat{P}_b = 1$ and the drain reaches zero: $n = 0$, and $z \leq 1$ by (iii); with $P_{\text{eff}} \geq L+1$ this gives $n + z \leq 1 \leq P_{\text{eff}} - L$, so $V \cap A^c = \emptyset$.

Proof of Cor. A.6. The reported persistence falls short by the history outside the run and what the run loses. The left inequality is Prop. A.1. For the right, (R6) gives $\widehat{P}_b \geq P_{\text{eff}} - n - z$ at t_{last} . On $A^c \cap V^c$ the count is moreover unwritten between t_{last} and t_d , so the value declared is the one at t_{last} : there $\widehat{P}_b \geq L$, so Line 14 skips a miss and Line 20 takes the report branch, leaving Lines 16 and 23 unreachable.